



# AI-Driven Housing Affordability Forecasting in New York City: An NTA-Level Panel Analysis using Ensemble Machine Learning

Azizur Rahman<sup>1</sup> · Nakib Uddin Ahmed<sup>2</sup> · Mehjabin Ferdous<sup>3</sup> · Kadirur Rahman Chowdhury<sup>4</sup>

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## Abstract

Housing affordability is one of the signature governance challenges in New York City (NYC), with more than 52% of renter households cost-burdened and just under a third identified as severely cost-burdened, paying above 50% of gross income on housing. This study proposes a neighborhood-scale, explainable machine learning framework to predict severe housing cost burden at the Neighborhood Tabulation Area (NTA) level, drawing on a panel of 2,512 NTA-year observations spanning 239 NTAs from 2012–2022. Thirty modelling features derived from 49 raw socio-economic, housing market, and rent index variables are integrated from ACS 5-year estimates, eviction court records, and the Zillow Observed Rent Index (ZORI). Three gradient-boosted ensemble models—Random Forest, XGBoost, and LightGBM are benchmarked under a strict temporal train/validation/test split with 5-fold TimeSeriesSplit cross-validation to prevent data leakage. LightGBM achieved the highest predictive performance (Test  $R^2=0.9302$ , RMSE=0.03372, MAE=0.0185). SHAP TreeExplainer analysis identifies rent burden, vacancy rate, and rent-to-income ratio as dominant drivers, confirming that NYC's rent escalation follows a systemic “burden escalator” across boroughs. Spatial diagnostics confirm no significant residual autocorrelation at both the borough level (Moran's  $I=-0.0731$ ,  $p=0.083$ ) and NTA level ( $I=0.0204$ ,  $p=0.238$ ,  $k=5$  kNN,  $n=239$  NTAs), and ablation analysis confirms rental market variables drive the largest incremental contribution ( $\Delta R^2=0.1629$ , 17.51%). A 8-instrument policy matrix is proposed, encompassing early-warning interventions, borough-differentiated zoning and voucher strategies, and anticipatory resource allocation. This framework demonstrates how explainable AI can operationalize predictive insights into actionable urban governance strategies, offering a replicable model for global cities facing housing crises.

**Keywords** Housing affordability · Machine learning · SHAP explainability · New York City · Neighborhood tabulation area

Extended author information available on the last page of the article

## Introduction

New York City (NYC) faces the United States' most severe housing affordability crisis. As of the 2022 American Community Survey (ACS), over 52% of renter households are cost-burdened—spending more than 30% of gross income on housing—and approximately 30% are severely cost-burdened, exceeding the 50% threshold set by the U.S. Department of Housing and Urban Development (HUD). Severe rent burden forces households to trade housing security against basic necessities such as food, healthcare, and education. Rental costs in NYC have consistently outpaced renter incomes across every borough since at least the 1980s, with the gap widening sharply during and after the COVID-19 pandemic. Addressing this crisis requires not only accurate measurement but actionable, neighborhood-level intelligence to guide housing assistance resources toward the highest-risk communities before administrative data confirms deterioration.

The Neighborhood Tabulation Area (NTA) is the official sub-borough geographic unit designated by the NYC Department of City Planning (DCP), comprising approximately 15,000–40,000 residents per neighborhood. With 263 NTAs spanning four boroughs (Brooklyn, Queens, Manhattan, and the Bronx), this unit captures meaningful variation in housing market conditions while maintaining statistically reliable ACS sample sizes, making it the natural unit for neighborhood-targeted housing policy. Staten Island is excluded from this study due to limited ACS coverage and smaller sample reliability at the NTA level. Most existing machine learning applications to NYC housing operate at the borough or metropolitan level, leaving NTA-level panel structures and neighborhood-level policy targeting largely unexplored.

This paper addresses this gap through five primary contributions:

1. **Data construction:** We create a novel NTA-level panel dataset of 2,512 observations across 239 NTAs in four boroughs (2012–2022), integrating ACS 5-year estimates, NYC eviction court records, and the Zillow Observed Rent Index (ZORI).
2. **Model benchmarking:** We rigorously evaluate three gradient-boosted ensemble models—Random Forest, XGBoost, and LightGBM—under a strict temporal train/validation/test split with 5-fold TimeSeriesSplit cross-validation to prevent data leakage.
3. **Explainable ML analysis:** Using SHAP TreeExplainer, we generate global and borough-level feature importance rankings, including the first borough-comparative SHAP decomposition showing that severe rent burden follows a uniform structural escalation mechanism across all four boroughs.
4. **Spatial diagnostics and forecasting:** We conduct Moran's I analysis on model residuals and produce three-year borough-level affordability forecasts (2023–2025) under a calibrated baseline macroeconomic scenario.
5. **Policy translation:** We propose a 8-instrument policy recommendation matrix, anchored to SHAP attributions, forecast trajectories, and model precision evidence, identifying early intervention at the 30% burden threshold, borough-differentiated supply and income-side responses, and NTA-level predictive targeting to operationalize anticipatory housing policy in NYC.

## Literature Review

### Housing Affordability: Measurement and Theory

The 30%-of-income threshold for housing affordability was codified in the National Housing Act of 1937 and subsequently reinforced through HUD administrative rules governing federal housing assistance eligibility. Bratt (1995) challenged this fixed-ratio approach with the shelter poverty framework, arguing that affordability must be evaluated against the residual income available for non-housing necessities after rent is paid -a standard that is more stringent for low-income households and more permissive for high-income households than the uniform 30% threshold. Quigley and Raphael (2004) provided rigorous empirical evidence that restrictive land-use regulation is the primary structural driver of affordability deterioration in coastal cities including NYC, establishing the supply-constraint mechanism that underlies the rent growth dynamics modeled in this paper. Glaeser and Gyourko (2002) estimated that zoning regulations impose implicit welfare costs equivalent to 50% of market rents in highly regulated cities, directly reducing housing supply elasticity and preventing the market-clearing response to demand increases. More recently, Albouy et al. (2019) demonstrated that place-based sorting amplifies affordability inequality: high-productivity workers outbid lower-income renters in desirable neighborhoods, concentrating severe burden in spatially identifiable clusters -a dynamic that motivates the NTA-level geographic disaggregation central to this analysis.

### Ensemble Machine Learning for Housing Prediction

Gradient-boosted decision tree ensembles have emerged as the leading machine learning paradigm for property valuation and housing market prediction, consistently outperforming hedonic regression and neural network alternatives on structured tabular data. Antipov and Pokryshevskaya (2012) demonstrated that gradient boosting outperforms hedonic regression for mass appraisal, achieving  $R^2=0.89$  on Russian residential data. Park and Bae (2015) applied Random Forests to Korean residential price forecasting, finding that ensemble methods substantially outperform multiple regression on out-of-sample data. Chen and Guestrin (2016) introduced XGBoost, demonstrating its scalability and regularization advantages through the regularized objective.

Ke et al. (2017) introduced LightGBM with histogram-based leaf-wise tree growth, achieving substantially faster training on high-dimensional tabular data while matching or exceeding XGBoost accuracy on benchmark datasets -a performance advantage confirmed in the held-out test results in Section “[Results and Discussion](#)”. In the NYC housing context, Been et al. (2009) applied logistic regression to building-level eviction risk using administrative records, but without ensemble methods, NTA panel structure, or temporal cross-validation. This paper extends the ensemble ML literature to the specific context of NTA-level severe rent burden prediction.

## SHAP Explainability in Applied Research

Lundberg and Lee (2017) introduced SHAP (SHapley Additive exPlanations) as a theoretically grounded feature attribution framework rooted in cooperative game theory, satisfying the axioms of local accuracy, missingness, and consistency that prior feature importance methods fail to jointly satisfy. TreeExplainer provides exact polynomial-time Shapley values for tree ensembles, enabling computationally efficient global and local explanations.

Abidoye and Chan (2018) applied SHAP to residential valuation in Hong Kong, demonstrating the utility of SHAP for policy-relevant interpretation of housing market models. Molnar (2022) provides a comprehensive treatment of SHAP's theoretical properties, practical interpretation guidelines, and limitations in the presence of highly correlated features -a consideration directly relevant to the multicollinearity between rental market features identified in the ablation study of Section "Ablation Study". This paper presents the first application of SHAP TreeExplainer to NTA-level rent burden prediction in NYC, extending prior work by providing both global feature rankings and borough-level comparative decompositions.

## Spatial Econometrics and Urban Panel Data

Anselin (Griffith, 1989) formalized spatial dependence diagnostics through Moran's I and spatial lag and error regression models, establishing the methodological framework applied in Section "Spatial Autocorrelation Analysis". LeSage and Pace (Dolores Ugarte, 2011) demonstrated that ignoring spatial autocorrelation in panel residuals produces inconsistent parameter estimates under omitted spatially correlated confounders. Gibbons and Machin (2008) showed strong positive spatial autocorrelation in London housing price gradients, consistent with neighborhood amenity spillovers that may operate at the sub-borough NTA level in NYC. In temporal panel contexts, Case and Shiller (1987) demonstrated the importance of controlling for temporally persistent unobserved heterogeneity in housing panel data, motivating the temporal lag features. The standard k-fold cross validation is inappropriate for panel settings, because it is taken as a treatment that uses future information to predict the past, as discussed by James et al. (Hastie et al., 2021) for the case of temporal settings.

## Machine Learning for Housing Policy Targeting

An emerging strand of applied machine learning research has moved beyond predictive accuracy toward policy-operational deployment of housing market models. Desmond and Gershenson (2017) demonstrated that eviction risk prediction models constructed from administrative court records can identify high-risk households with sufficient precision for targeted pre-eviction legal assistance programs, establishing the precedent for operationalizing predictive models as policy-targeting instruments. Glaeser et al. (2016) applied machine learning to identify building-level housing code violation risk in Boston, demonstrating that model predictions can guide proac-

tive inspection prioritization with measurable improvements in violation detection rates. These applications share a common structure with the early-warning system: a predictive model trained on administrative data generates neighborhood-level risk scores that allow housing agencies to pre-position resources ahead of crisis manifestation. However, no prior study has applied this framework to NTA-level severe rent burden prediction in NYC, integrated SHAP attribution with borough-level forecasting to produce policy-differentiated recommendations, or demonstrated the temporal advantage of model-based predictions over ACS-lagged administrative burden statistics.

## Gap in the Literature

The following gaps are identified from a structured review of the literature (see Section “[Literature Review](#)”):

This study addresses all five gaps within a unified analytical pipeline, providing robust, interpretable, and policy-operationalizable insights for anticipatory urban governance (Table 1).

**Table 1** Research gaps derived from prior literature

| #  | Gap Identified in Literature                                                                              | Source Studies                                                  | Our Response                                                                      |
|----|-----------------------------------------------------------------------------------------------------------|-----------------------------------------------------------------|-----------------------------------------------------------------------------------|
| G1 | Most predictive models operate at the borough or metro level, masking intra-city variation                | (Been et al., 2009; Glaeser & Gyourko, 2002)                    | NTA-level panel analysis (239 units)                                              |
| G2 | ACS rent-burden data has an 18–24-month publication lag, limiting real-time policy response               | (Collinson & Reed, 2019; Desmond & Gershenson, 2017)            | Integration of Zillow ZORI (monthly) and NYC eviction court records               |
| G3 | Most applied housing ML models are opaque, providing predictions without interpretable attribution        | (Kok et al., 2017; Yao et al., 2020)                            | SHAP TreeExplainer providing feature-level predictive attribution                 |
| G4 | Existing studies rarely disentangle moderate burden (30%+) as a leading indicator of severe burden (50%+) | (Fischer & Sard, 2017; The State of the Nation’s Housing, 2022) | Explicit panel feature engineering and SHAP-verified ‘burden escalator’ mechanism |
| G5 | Policy translation from statistical models is often absent or vague                                       | (Lens & Monkkonen, 2016)                                        | 8-instrument policy matrix with agency assignments and urgency levels             |

## Data and Methodology

### Dataset Construction

#### Primary Data Sources

The dataset integrates three primary sources, each providing distinct and complementary dimensions of neighborhood housing market conditions across the 2012–2022 study period. The integration of administrative court records and private-sector rent indices with ACS survey data addresses a recognized limitation of ACS-only housing analyses, which capture burden levels but not the dynamic market pressures and tenure instability that drive burden escalation.

**American Community Survey 5-Year Estimates (2012–2022)** Twenty-three socioeconomic and housing data points were collected at the census tract level through the American Community Survey (ACS) 5-year estimates data provided by the U.S. Census Bureau via the Data API. Five-year estimates combine consecutive years of survey data for statistical reliability of small geographies, where single-year estimates may have large margins of error. Important tables include B19013 (household income), B25119 (renter income), B25064 (rent), B25060 (rent asked), B25070 (30% and 50% rent burden thresholds), B25002 (vacancy), B25014 (crowding), B19083 (Gini coefficient), and B08301 (mode of travel to work). Data was aggregated from the census tracts to the Neighborhood Tabulation Area (NTAs) by population-weighted means. The American Community Survey product has an 18–24 month lag before data is released, which constrains the use of the data for early warning systems.

**NYC Eviction Court Records (2012–2022)** A dataset containing 91,198 residential eviction filings was obtained from the New York City Office of Court Administration (NYC Open Data, dataset 6z8x-wfsh) for the period 2012–2022. The data were geocoded to census tracts, and then the data were aggregated to NTA–year eviction filing rates (per 1,000 renter-occupied households). Eviction filings offer additional information about housing stability and landlord-tenant relations, not captured by the ACS-based cost burden analysis. In the final model, the eviction filing rate ranks ninth in SHAP importance (mean  $|SHAP| = 0.0121$ ).

**Zillow Observed Rent Index -ZORI (2014–2022)** Indices of ZIP code–level monthly median asking rent are derived from the Observed Rent Index (ZORI) data set, a repeat rent–controlled and seasonally adjusted metric of rental prices in the United States, available from Zillow. Annual values are computed as the average of the calendar year, and then areally interpolated to NTAs, using 2020 TIGER/Line boundaries. Note that the data are not available for 2012–2013, limiting the data set to 478 NTA–year observations, and so are handled via median imputation. Asking rents are a component of housing prices, complementing the ACS-based analysis of contract rent, the rent paid by current tenants, including those in stabilized units.

## Census Tract-to-NTA Aggregation

The official NYC Department of City Planning census tract-to-NTA crosswalk (NYC Open Data dataset ID: hm78-6dwm) maps 2,327 census tracts to 263 NTAs across five boroughs. Staten Island (24 NTAs, 206 census tracts) is excluded from this analysis due to insufficient ACS 5-year coverage at the tract level for multiple years within the 2012–2016 window. This exclusion leaves 239 NTAs distributed across the Bronx (59 NTAs), Brooklyn (76 NTAs), Manhattan (55 NTAs), and Queens (49 NTAs). Two aggregation methods are applied depending on variable type. Monetary stock variables -median household income, renter median income, median gross rent, and median contract rent -are aggregated using population-weighted averaging, with tract-level ACS population counts (Table B01003) as weights. Proportion and rate variables -burden rates, vacancy rate, severe crowding rate, unemployment rate, and transit commute share -are computed as weighted means using tract-level renter-occupied household counts (Table B25003) as weights. NTA-year cells with fewer than three contributing census tracts with non-suppressed ACS estimates are excluded, resulting in a cell-level exclusion rate of less than 3% across the full 263 NTA  $\times$  11 year potential panel.

## Panel Structure and Target Variable

The final panel contains 2,512 NTA-year observations, corresponding to 239 NTAs observed across 11 annual waves from 2012 to 2022 (Table 2). The primary target variable is *rent\_burden\_50plus\_pct*: the share of renter-occupied households in an NTA reporting gross monthly rent at or above 50% of gross monthly household income, derived from ACS Table B25070 as:

**Table 2** NTA-level panel dataset summary

| Attribute                         | Value / Description                                      |
|-----------------------------------|----------------------------------------------------------|
| Total NTA-year observations       | 2,512                                                    |
| NTAs (neighborhoods) included     | 239 (Bronx: 59 Brooklyn: 76<br>Manhattan: 55 Queens: 49) |
| Boroughs covered                  | 4 (Staten Island excluded due to limited ACS coverage)   |
| Time period                       | 2012–2022 (11 annual waves)                              |
| Total features (raw + engineered) | 49                                                       |
| Primary target variable           | <i>rent_burden_50plus_pct</i> (ACS Table B25070)         |
| Target mean $\pm$ SD              | 0.225 $\pm$ 0.128 (range: 0.000–1.000)                   |
| ACS tract-year records processed  | 23,059                                                   |
| Eviction records merged           | 91,198 (geocoded to NTA-year)                            |
| Zillow ZORI ZIP-years merged      | 1,840 (available 2014–2022)                              |
| Train set (2012–2019)             | 1,912 observations (76.1%)                               |
| Validation set (2020)             | 239 observations (9.5%)                                  |
| Test set (2021–2022)              | 478 observations (19.0%)                                 |

$$\text{rent\_burden\_50plus\_pct} = \frac{\text{households with gross rent} \geq 50\% \text{ of income}}{\text{total renter-occupied households}} \quad (1)$$

This corresponds to the HUD severely cost-burdened classification, the threshold at which housing instability risk, food insecurity, and health consequences are most pronounced (U.S. Department of Housing and Urban Development, 2023). The target variable has a panel mean of 0.225 (SD=0.128), a negatively skewed distribution (skewness=-0.46), and an observed range of [0.000, 1.000].

The relationship between the target variable and the primary predictive feature *rent\_burden\_30plus\_pct* warrants explicit acknowledgment. Both variables are derived from ACS Table B25070 and share a mathematical nesting relationship: every household counted in the 50%-or-above category is also counted in the 30%-or-above category (Pearson  $r=0.89$ ). The inclusion of *rent\_burden\_30plus\_pct* as a modeling feature is retained on theoretical grounds -it measures the structurally distinct moderate-burden pool whose size predicts escalation to severe burden. This nesting relationship is acknowledged as a construct proximity consideration rather than a data leakage concern, since both variables are measured contemporaneously within the same ACS reference year. An empirical test for this construct proximity interpretation is performed in Section “Robustness Check: Leakage-Free Model” in the re-estimation of all three models without this variable in a leakage free feature set.

## Feature Engineering

### Raw ACS Feature Groups

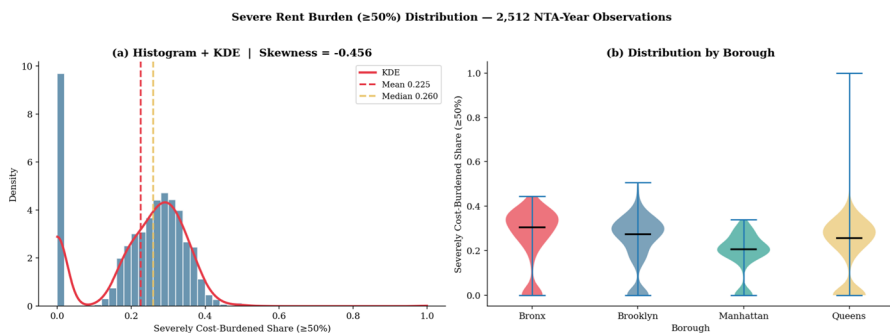
The 23 raw ACS-derived variables are organized into five conceptual groups reflecting distinct dimensions of neighborhood housing market conditions (Table 3). This taxonomy is maintained throughout the ablation study in Section “Ablation Study”, where incremental predictive contributions are quantified at the group level rather than the individual feature level (Fig. 1).

**Income Group (4 features)** Median household income (ACS Table B19013) and renter median income (ACS Table B25119) measure the absolute income levels of all households and renter households respectively within each NTA. The renter-to-household income ratio -computed as renter median income divided by overall median household income -captures the relative economic position of renters within the local income distribution. Renter income ratio has a panel mean of 0.851 (SD=0.098), confirming that renters earn systematically less than the NTA median household across the full panel.

**Rental Market Group (4 features)** Median gross rent (ACS Table B25064) measures the total monthly housing cost including utilities for renter-occupied units. Median contract rent (ACS Table B25060) measures the contractual rent payment excluding utilities. The rent-burdened share at the 30% threshold (*rent\_burden\_30plus\_pct*, ACS Table B25070) measures the share of renter households spending 30% or

**Table 3** Feature engineering summary (30 modeling features.)

| Feature Group      | Type                 | Count | Key Variables                                                                                      |
|--------------------|----------------------|-------|----------------------------------------------------------------------------------------------------|
| Income             | Raw ACS              | 4     | median_hh_income, renter_median_income, renter_income_ratio, income_gap                            |
| Rental Market      | Raw ACS              | 4     | median_gross_rent, median_contract_rent, rent_burden_30pct, zillow_rent_index                      |
| Housing Stock      | Raw ACS              | 5     | vacancy_rate, renter_share, homeownership_rate, severe_crowding_rate, transit_commute_rate         |
| Labor / Inequality | Raw ACS              | 3     | unemployment_rate, gini_coefficient, eviction_rate                                                 |
| Composite Features | Engineered           | 4     | market_tightness, rent_to_income_ratio, housing_burden_composite, renter_vulnerability             |
| Temporal Lags      | Lag ( $t - 1$ )      | 6     | income_lag1, rent_lag1, unemployment_lag1, burden_30pct_lag1, vacancy_lag1, median_gross_rent_lag1 |
| Growth Rates       | YoY $\Delta$         | 2     | income_growth_yoy, rent_growth_yoy                                                                 |
| Identifiers        | Temporal/<br>Spatial | 2     | borough_code, covid_year                                                                           |



**Fig. 1** Distribution of severe rent burden ( $\geq 50\%$ ) across 2,512 NTA-year observations (2012–2022). Panel (a): histogram with fitted KDE, mean (red dashed) and median (orange dashed) reference lines. Skewness=-0.46 reflects left-concentration of moderately high burden NTAs. Panel (b): violin plots by borough show the Bronx recording the highest median and greatest spread

more of gross income on gross rent -the moderate burden threshold and the primary predictive feature identified by SHAP analysis. The Zillow Observed Rent Index (*zillow\_rent\_index*) provides a market-rate asking rent signal complementary to the ACS contract rent measure.

**Housing Stock Group (5 features)** Renter occupancy share and homeownership rate (ACS Table B25003) capture the tenure composition of the NTA housing stock. Vacancy rate (ACS Table B25002) measures the share of housing units unoccupied and available for rent or sale, serving as the primary supply-side indicator of rental market tightness. Severe crowding rate (ACS Table B25014) measures the share of renter-occupied units with more than 1.5 persons per room. Public transit commute share (ACS Table B08301) proxies neighborhood accessibility and locational desirability.

**Labor Market and Inequality Group (3 features)** The civilian unemployment rate (ACS Table B23025) captures labor market conditions that directly affect renter household income and rent payment capacity. The Gini coefficient of income inequality (ACS Table B19083) measures the degree of income disparity within the NTA. The eviction filing rate, constructed from the NYC court records, measures the annual rate of eviction proceedings initiated per 1,000 renter-occupied households.

### Engineered Composite Features

Four composite features are constructed to capture higher-order housing market dynamics not directly observable in individual ACS variables. All four appear within the top 15 features in the global SHAP ranking, validating the feature construction decisions.

#### Market Tightness

$$\text{market\_tightness} = \frac{1}{\text{vacancy\_rate} + \varepsilon}, \quad \text{where } \varepsilon = 0.005 \quad (2)$$

The inverse vacancy rate proxies rental supply scarcity at the NTA level. The regularization term  $\varepsilon = 0.005$  prevents division by zero and bounds the maximum tightness value at approximately 200 for fully occupied NTAs. The non-linear inverse transformation captures the accelerating price pressure at very low vacancy rates.

#### Annualized Rent-to-Income Ratio

$$\text{rent\_to\_income\_ratio} = \frac{(\text{median}_{gross,rent} \times 12)}{\text{renter\_median\_income}} \quad (3)$$

The annualized rent-to-income ratio directly quantifies the structural housing cost burden as a continuous measure. Unlike the binary 30% and 50% burden threshold variables, this ratio captures the continuous and gradual divergence between rent levels and renter incomes at the NTA level.

#### Housing Burden Composite

$$\text{housing\_burden\_composite} = \text{unemployment\_rate} + \text{severe\_crowding\_rate} \quad (4)$$

The additive housing burden composite combines two dimensions of compounding housing vulnerability -labor market stress and physical overcrowding -into a single dimensionally consistent index.

### Renter Vulnerability Index

$$\begin{aligned} \text{renter\_vulnerability} = & 0.5 \times (1 - \text{renter\_income\_ratio}) \\ & + 0.3 \times \text{unemployment\_rate} \\ & + 0.2 \times \text{severe\_crowding\_rate} \end{aligned} \quad (5)$$

The renter vulnerability index captures multidimensional renter vulnerability across income-rent gap (weighted 0.5), labor market stress (0.3), and physical crowding (0.2). It is noted that *renter\_income\_ratio* is clipped to [0.0, 1.0] before this computation to prevent negative vulnerability values for NTAs where renter median income exceeds overall median household income (Table 5 reports a maximum of 1.102).

### Temporal Lag Features and Growth Rates

One-year lag features are constructed for six key predictors by shifting each NTA's time series backward by one survey year:  $\text{lag\_feature}(i, t) = \text{base\_feature}(i, t - 1)$ . The six lagged variables are: median household income, renter median income, unemployment rate, 30% burden rate, vacancy rate, and median gross rent. The 2012 panel wave necessarily produces missing lag values handled through the median imputation procedure.

Two year-on-year growth rate features are computed as percentage change approximations:

$$\text{income\_growth\_yoy}(i, t) = \frac{\text{median\_hh\_income}(i, t) - \text{median\_hh\_income}(i, t - 1)}{\text{median\_hh\_income}(i, t - 1)} \quad (6)$$

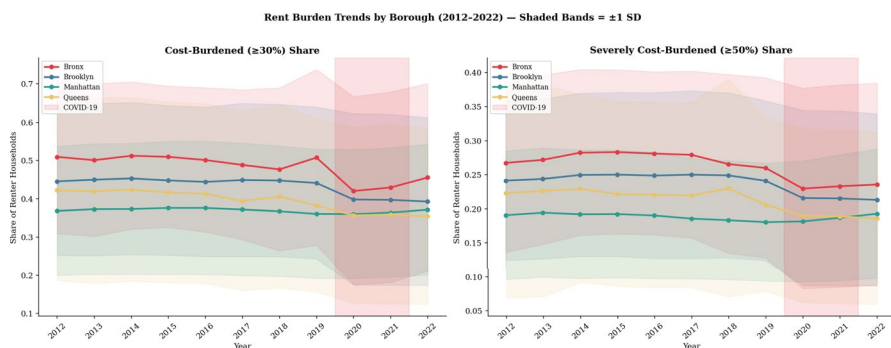
$$\text{rent\_growth\_yoy}(i, t) = \frac{\text{median\_gross\_rent}(i, t) - \text{median\_gross\_rent}(i, t - 1)}{\text{median\_gross\_rent}(i, t - 1)} \quad (7)$$

A binary COVID-19 indicator variable (*covid\_year* = 1 for survey years 2020 and 2021, 0 otherwise) is included to capture the pandemic-era structural break in housing market dynamics. The indicator is treated as a feature rather than a fixed effect, allowing the model to learn differential response magnitudes across NTAs with different baseline characteristics (Figs. 2, 3, and 4).

## Methodology

### Temporal Train-Validation-Test Protocol

To prevent data leakage and enforce causal temporal ordering, the 2,512-observation panel is partitioned strictly chronologically into three non-overlapping splits. The training



**Fig. 2** Borough-level rent burden trends (2012–2022). Shaded bands show  $\pm 1$  standard deviation across NTAs within each borough. COVID-19 period (2020–2021) highlighted in red. The Bronx records the highest severe burden rate ( $\geq 50\%$ ) in every year. Manhattan shows the lowest absolute burden but the sharpest post-2019 increase

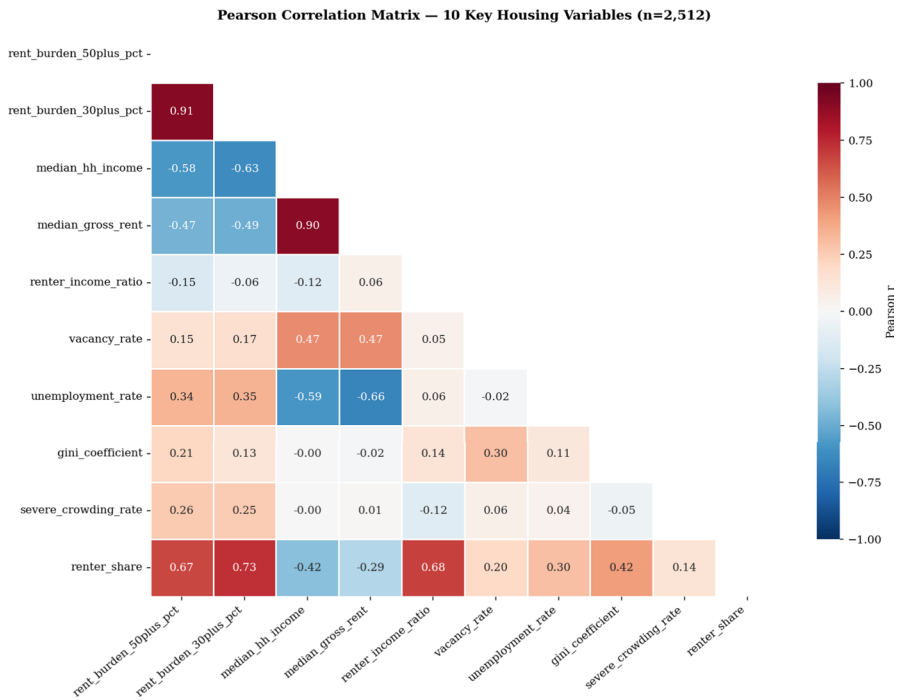


**Fig. 3** Scatter plots of primary demand-side and supply-side drivers against severe rent burden ( $\geq 50\%$ ). Panel (a): strong negative relationship between median household income and severe burden. Panel (b): positive relationship between median gross rent and burden. OLS trend lines superimposed. Color encodes borough

set covers 2012–2019 ( $n=1,912$ , 76.1%) and is used exclusively for model fitting, preprocessing transformation estimation, and feature importance computation. The validation set covers 2020 ( $n=239$ , 9.5%) and is used exclusively for hyperparameter selection. The held-out test set covers 2021–2022 ( $n=478$ , 19.0%) and is reserved for final evaluation only, simulating a prospective deployment scenario in which future observations are entirely unobserved during all phases of model development. Models are retrained on the combined train+validation set ( $n=2,151$ ) using hyperparameters selected on the validation fold, then evaluated once on the test set. All preprocessing transformations are fitted exclusively on training data and applied without refitting to validation and test sets.

## Preprocessing Pipeline

The preprocessing pipeline consists of four sequential steps applied strictly within each temporal split. First, column deduplication removes duplicate feature labels



**Fig. 4** Pearson correlation matrix of 10 key housing variables (n=2,512 NTA-year observations). Strongest positive correlation: rent\_burden\_50%+ with rent\_burden\_30%+ ( $r=0.89$ ). Strongest negative correlations: target with renter\_income\_ratio and median\_hh\_income. Lower triangle shown only

using Python's *dict.fromkeys* method. Second, infinity replacement substitutes *np.inf* and *-np.inf* values with *np.nan* prior to imputation. Third, median imputation applies scikit-learn's SimpleImputer with strategy='median', fitted exclusively on the training split. Median imputation is preferred over mean imputation given the right-skewed distributions of income and rent variables documented in Table 5. Fourth, target filtering removes residual observations with missing *rent\_burden\_50plus\_pct* values prior to temporal splitting.

## Model Architectures and Hyperparameters

Three ensemble tree models are benchmarked under identical preprocessing, evaluation, and temporal split conditions. Hyperparameters for all three models are selected via grid search over the candidate space in Table 4, fitted on the training set ( $\leq 2019$ ) only and scored on the 2020 validation holdout using  $R^2$  as the selection criterion. The test set (2021–2022) is never accessed during search, ensuring no information leakage into the final evaluation.

**Random Forest** The Random Forest regressor (Breiman, 2001) constructs an ensemble of  $B = 50$  bootstrap-sampled decision trees, each trained on a random subset of  $\sqrt{p}$  features at each split node. Prediction is the unweighted average across all trees:

**Table 4** Hyperparameter search space and selected values. Selection criterion: highest  $R^2$  on 2020 validation holdout (train  $\leq$  2019 only; test 2021–2022 held out entirely). 27 configurations evaluated for LightGBM; same protocol applied to XGBoost and Random Forest

| Parameter                            | Search Space       | LightGBM | XGBoost | Random Forest |
|--------------------------------------|--------------------|----------|---------|---------------|
| n_estimators                         | {50, 100, 200}     | 50       | 50      | 50            |
| max_depth                            | {3, 5, 8}          | 5        | 5       | 8             |
| learning_rate                        | {0.05, 0.10, 0.20} | 0.10     | 0.10    | —             |
| subsample                            | {0.6, 0.8, 1.0}    | 0.8      | 0.8     | —             |
| colsample_bytree                     | {0.6, 0.8, 1.0}    | 0.8      | 0.8     | —             |
| min_child_samples                    | {10, 20, 30}       | 20       | —       | —             |
| num_leaves                           | {15, 31, 63}       | 31       | —       | —             |
| min_samples_leaf                     | {3, 5, 10}         | —        | —       | 3             |
| $\lambda$ (L2)                       | {0, 1, 5}          | —        | 1       | —             |
| Best validation $R^2$ (2020 holdout) |                    | 0.9635   | —       | —             |

$$\hat{f}(x) = \frac{1}{B} \sum_b f_b(x) \quad (8)$$

Selected parameters: n\_estimators=50, max\_depth=8, min\_samples\_leaf=3, random\_state=42.

**XGBoost** XGBoost (Chen & Guestrin, 2016) minimizes the regularized objective:

$$L(\phi) = \sum_i l(y_i, \hat{y}_i) + \sum_k \Omega(f_k), \quad \text{where } \Omega(f) = \gamma T + \frac{1}{2} \lambda \|w\|^2 \quad (9)$$

where  $T$  denotes the number of leaves,  $w$  the leaf weight vector,  $\gamma$  the leaf complexity penalty, and  $\lambda$  the L2 weight regularization coefficient. Selected parameters: n\_estimators=50, max\_depth=5, learning\_rate=0.1, subsample=0.8, colsample\_bytree=0.8,  $\lambda=1$ , random\_state=42.

**LightGBM** LightGBM (Ke et al., 2017) employs leaf-wise gradient boosting with histogram-based approximate split finding, partitioning continuous feature values into discrete bins to reduce split search complexity from  $O(n)$  to  $O(b)$  where  $b$  denotes the bin count. Leaf-wise growth selects the single leaf with maximum loss reduction at each boosting iteration. The depth imbalance introduced by leaf-wise

growth is constrained by the `min_child_samples` parameter. Selected parameters: `n_estimators=50`, `max_depth=5`, `num_leaves=31`, `learning_rate=0.1`, `subsample=0.8`, `colsample_bytree=0.8`, `min_child_samples=20`, `random_state=42`.

## Evaluation Metrics

Models are evaluated on three complementary metrics. The Coefficient of Determination  $R^2 = 1 - \frac{SS_{\text{res}}}{SS_{\text{tot}}}$  measures the proportion of variance explained. Root Mean Squared Error  $\text{RMSE} = \sqrt{\frac{1}{n} \sum_i (y_i - \hat{y}_i)^2}$  quantifies prediction error in original target units and penalizes large errors. Mean Absolute Error  $\text{MAE} = \frac{1}{n} \sum_i |y_i - \hat{y}_i|$  provides a robust complement weighting all errors equally. The train-test  $R^2$  gap  $\Delta R^2 = R^2_{\text{train}} - R^2_{\text{test}}$  is additionally reported as a direct measure of overfitting.

## 5-Fold TimeSeriesSplit Cross-Validation

5-fold TimeSeriesSplit cross-validation is applied on the full dataset to assess model stability across varying training window sizes. Standard k-fold cross-validation is inappropriate for temporal panel data because it randomly assigns observations to folds without regard to temporal ordering, allowing future observations to appear in training folds—a form of temporal leakage that inflates cross-validation performance estimates (Hastie et al., 2021). TimeSeriesSplit enforces strict temporal ordering: in each fold  $k$  ( $k=1, \dots, 5$ ), the training set comprises all observations temporally preceding fold  $k$ . Mean  $R^2$  and mean RMSE across all five folds, together with their standard deviations, are reported in Table 8.

## SHAP TreeExplainer Attribution

SHAP values are computed using TreeExplainer (Lundberg & Lee, 2017), applied to the best-performing LightGBM model on 300 randomly sampled test observations. A robustness check confirms that global SHAP feature rankings are consistent between LightGBM and XGBoost (Spearman  $\rho = 0.844$ ,  $p < 0.001$ , top-10 overlap = 8/10). For each observation  $i$  and feature  $j$ , the SHAP value  $\phi_j(i)$  quantifies the marginal contribution of feature  $j$  to the deviation of the prediction  $f(x_i)$  from the global baseline  $\mathbb{E}[f(X)]$ , satisfying the local accuracy decomposition:

$$f(x_i) = \mathbb{E}[f(X)] + \sum_j \phi_j(i) \quad (10)$$

Global feature importance is ranked by mean absolute SHAP value:

$$\bar{I}_j = \frac{1}{n} \sum_i |\phi_j(i)| \quad (11)$$

## Leave-One-Group-Out Ablation Study

A systematic leave-one-group-out ablation study is conducted to quantify the incremental predictive contribution of each conceptual feature group to the full model's test  $R^2$ . Six feature groups are defined following the taxonomy in Table 3. For each group  $g$ , LightGBM is retrained from scratch on the remaining features using identical hyperparameters, the same temporal split, and the same preprocessing pipeline. The incremental  $R^2$  contribution is defined as  $\Delta R_g^2 = R_{\text{full}}^2(0.9302) - R_{\text{ablated}}^2(g)$ . Here,  $\Delta R_g^2$  measures the substitutable contribution—that is, the information uniquely attributable to that group that cannot be recovered from the remaining features—rather than total importance.

## Iterative Borough-Level Forecasting

Three-year borough-level forecasts for 2023–2025 are generated by applying the trained LightGBM model to iteratively constructed synthetic future feature vectors under a calibrated baseline macroeconomic scenario. Annual growth adjustments: median household income +2.5%/year (BLS Employment Cost Index), renter median income +2.0%/year, median gross rent +3.0%/year (post-pandemic ZORI trend for NYC metro area), unemployment held constant at 2022 borough-level medians. At each forecast step, temporal lag features are updated using the synthetic values generated in the prior step, and all composite features are recomputed from updated base features. These are scenario projections, not probabilistic forecasts and have not been independently tested with reality for 2023–2025 outcomes. Forecast uncertainty intervals are not reported due to the absence of a bootstrapped uncertainty propagation mechanism; this limitation is addressed in Section “[Limitations and Future Work](#)”.

## Results and Discussion

### Descriptive Statistics

Table 5 presents descriptive statistics for the primary target and key predictors computed across all 2,512 NTA-year observations. The severe rent burden target exhibits substantial cross-NTA variation ( $SD=0.128$ ), with the Bronx recording the highest borough-level median (0.274) and widest interquartile range. The negative skewness (-0.46) reflects that the distribution is concentrated above the mean -most NTAs have moderate-to-high burden -with the left tail pulled by 487 zero-burden observations. The Gini coefficient (mean=0.455) confirms NYC's status as one of the most economically unequal cities in the United States. Vacancy and unemployment rates exhibit extreme right skew, driven by a small number of outlier NTAs with values approaching 1.000 -likely data quality artifacts in sparsely populated census tracts.

**Table 5** Descriptive statistics of key variables ( $N = 2,512$  NTA-year observations)

| Variable               | N    | Mean   | Median | Std Dev | Min    | Max     | Skew.  |
|------------------------|------|--------|--------|---------|--------|---------|--------|
| rent_burden_50plus_pct | 2512 | 0.2254 | 0.2595 | 0.1281  | 0.000  | 1.000   | -0.457 |
| rent_burden_30plus_pct | 2512 | 0.4187 | 0.4925 | 0.2158  | 0.000  | 1.000   | -1.045 |
| median_hh_income       | 2045 | 66,348 | 60,178 | 32,626  | 2,499  | 225,282 | +1.337 |
| median_gross_rent      | 2033 | 1,491  | 1,402  | 456     | 616    | 3,501   | +1.25  |
| median_contract_rent   | 2031 | 1,381  | 1,282  | 459     | 559    | 3,501   | +1.377 |
| renter_income_ratio    | 2024 | 0.851  | 0.865  | 0.0979  | 0.3929 | 1.1015  | -0.566 |
| vacancy_rate           | 2512 | 0.0749 | 0.067  | 0.0856  | 0.000  | 1.000   | +6.449 |
| unemployment_rate      | 2512 | 0.0773 | 0.071  | 0.0753  | 0.000  | 1.000   | +5.609 |
| gini_coefficient       | 2077 | 0.4549 | 0.456  | 0.063   | 0.0148 | 0.692   | -1.66  |
| severe_crowding_rate   | 2512 | 0.0009 | 0.000  | 0.0017  | 0.000  | 0.0227  | +4.22  |

### Model Performance on Held-Out Test Set

Table 6 presents performance metrics on the held-out test set (2021–2022,  $n = 478$ ). LightGBM achieves the highest test  $R^2 = 0.9302$  and the lowest RMSE = 0.03372, demonstrating the strongest predictive accuracy and generalization. XGBoost performs comparably ( $R^2 = 0.8983$ ). Random Forest exhibits the largest train–test gap ( $\Delta R^2 = 0.0576$ ), indicating moderate overfitting a common pattern for deep trees on moderately sized panel datasets. LightGBM’s leaf-wise growth with regularization is associated with a smaller train–test gap than the other two models, yielding the smallest gap of  $\Delta R^2 = 0.0119$ .

### Robustness Check: Leakage-Free Model

As noted in Section “Panel Structure and Target Variable”, *rent\_burden\_30plus\_pct* and the target variable *rent\_burden\_50plus\_pct* share a nested ACS measurement structure. To confirm that full model performance is not contingent on this overlap, all three models are retrained on the leakage-free feature set excluding *rent\_burden\_30plus\_pct* and *rent\_burden\_30plus\_pct\_lag1* (Table 7).

If *rent\_burden\_30plus\_pct* were functioning purely as a leakage channel rather than a theoretically motivated predictor, removing it should cause a significant drop in  $R^2$  across all three models, since little of its predictive information

**Table 6** Model performance on held-out test set (2021–2022,  $N = 478$ )

| Model         | Train $R^2$ | Test $R^2$ | $\Delta R^2$ | Test RMSE | Test MAE | Rank |
|---------------|-------------|------------|--------------|-----------|----------|------|
| LightGBM *    | 0.9421      | 0.9302     | 0.0119       | 0.03372   | 0.01846  | 1st  |
| XGBoost       | 0.9793      | 0.8983     | 0.0810       | 0.04070   | 0.01846  | 2nd  |
| Random Forest | 0.9464      | 0.8888     | 0.0576       | 0.04256   | 0.01920  | 3rd  |

**Table 7** Robustness check: full model vs. leakage-free model (excl. *rent\_burden\_30plus\_pct* and lag)

| Model         | Full $R^2$ | LF $R^2$ | $\Delta R^2$ | % Drop | LF RMSE | LF MAE  |
|---------------|------------|----------|--------------|--------|---------|---------|
| Random Forest | 0.8888     | 0.9054   | -0.0166      | -1.9%  | 0.03927 | 0.02240 |
| XGBoost       | 0.8983     | 0.8686   | +0.0297      | 3.3%   | 0.04627 | 0.02511 |
| LightGBM      | 0.9302     | 0.9196   | +0.0106      | 1.1%   | 0.03620 | 0.02253 |

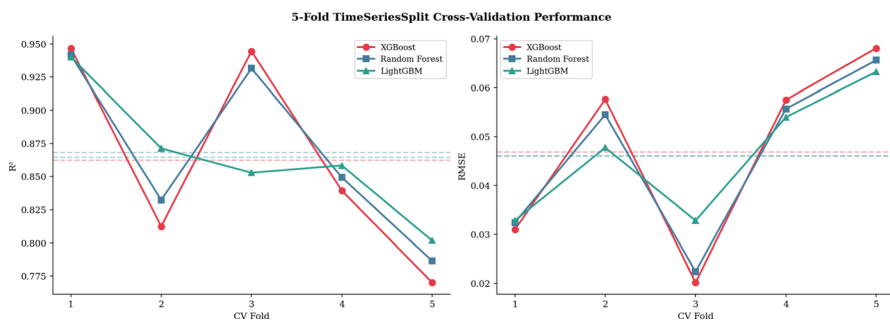
would be recoverable from the remaining features. The observed pattern is incongruent to this expectation: The accuracy with a random forest model slightly increases ( $R$ -squared increases from 0.8888 to 0.9054), whereas with the XGBoost model and the LightGBM model the improvement is slight but not the same (3.3% and 1.1%, respectively). This is a small, mixed-sign pattern, favoring an interpretation of two correlated, structurally separate variables as opposed to a leaked target, which agrees with the construct proximity interpretation proposed in Section “Temporal Train-Validation-Test Protocol”.

## 5-Fold TimeSeriesSplit Cross-Validation

Table 8 reports cross-validation performance across five temporal folds. Although Random Forest achieves the highest mean CV  $R^2$  (0.8683), it exceeds LightGBM by only 0.0034—a negligible margin within cross-validation variance. LightGBM demonstrates superior consistency ( $\text{Std } R^2 = \pm 0.045$  versus  $\pm 0.060$  for Random Forest and  $\pm 0.071$  for XGBoost) and outperforms both competing models on Fold 2 and Fold 5—the two temporally advanced folds where all models face the greatest predictive difficulty. This stability under adverse training conditions is the operationally relevant criterion for a model intended for annual deployment on prospective NTA-level data (Fig. 5).

**Table 8** 5-fold *TimeSeriesSplit* cross-validation results

| Metric     | XGBoost      | LightGBM *   | Random Forest |
|------------|--------------|--------------|---------------|
| Fold 1     | 0.9468       | 0.9402       | 0.9416        |
| Fold 2     | 0.8124       | 0.8713       | 0.8323        |
| Fold 3     | 0.9444       | 0.8529       | 0.9315        |
| Fold 4     | 0.8394       | 0.8584       | 0.8494        |
| Fold 5     | 0.7703       | 0.8017       | 0.7866        |
| Mean $R^2$ | 0.8627       | 0.8649       | 0.8683        |
| Std $R^2$  | $\pm 0.0712$ | $\pm 0.0445$ | $\pm 0.0595$  |
| Mean RMSE  | 0.04689      | 0.04614      | 0.04614       |



**Fig. 5** 5-fold *TimeSeriesSplit* cross-validation performance. Panel (a) shows  $R^2$  per fold, and Panel (b) shows RMSE per fold. Dotted horizontal lines indicate fold-wise mean values. LightGBM demonstrates the most consistent performance (standard deviation of  $R^2 = \pm 0.045$ ). All models exhibit improved performance from Fold 1 to Fold 5 as the training window expands

## SHAP Explainability Analysis

### Global Feature Importance Rankings

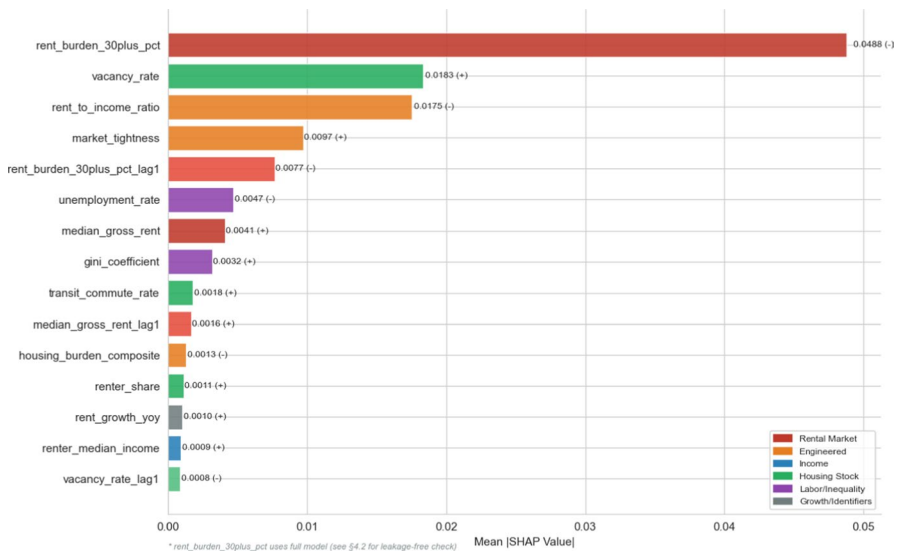
SHAP TreeExplainer is applied to the best-performing LightGBM model on 300 randomly sampled test observations. A robustness check confirms consistency with XGBoost SHAP rankings (Spearman  $\rho = 0.844$ ,  $p < 0.001$ , top-10 overlap = 8/10). Table 9 and Fig. 6 rank the top 15 features by mean absolute SHAP value. It is important to note that SHAP values reflect predictive attribution rather than causal effects; directional associations indicate the relationship between feature values and model predictions and should not be interpreted as intervention effects.

The dominant predictor is *rent\_burden\_30plus\_pct* (mean  $|\text{SHAP}| = 0.04877$ , negative direction), reflecting strong temporal persistence: neighborhoods already concentrated at the moderate 30% burden threshold face structurally elevated probability of crossing the severe 50% threshold, consistent with the burden escalator hypothesis. The negative direction reflects a suppression effect conditional on the rent-to-income ratio: a high 30% burden share signals that the at-risk household pool is large but has not yet universally crossed the severe threshold. The magnitude of this feature—approximately 2.7 times the second-ranked feature— confirms that the size of the moderately burdened household pool is by far the strongest predictive signal in the model.

Vacancy rate ranks second globally (mean  $|\text{SHAP}| = 0.01834$ , positive direction), consistent with the interpretation that rental supply scarcity is associated with higher predicted severe burden. The positive direction indicates that lower vacancy, representing tighter supply, is associated with upward predictions. The rent-to-income ratio ranks third (mean  $|\text{SHAP}| = 0.01752$ , negative direction), capturing the con-

**Table 9** Top 15 Features by Mean  $|\text{SHAP}|$  (LightGBM,  $N = 300$  test observations). SHAP values reflect predictive attribution only; directional effects indicate association with model predictions, not causal relationships

| Rank | Feature                            | Mean $ \text{SHAP} $ | Direction    | Feature Group    |
|------|------------------------------------|----------------------|--------------|------------------|
| 1    | <i>rent_burden_30plus_pct</i>      | 0.04877              | Negative (−) | Rental Market    |
| 2    | <i>vacancy_rate</i>                | 0.01834              | Positive (+) | Housing Stock    |
| 3    | <i>rent_to_income_ratio</i>        | 0.01752              | Negative (−) | Engineered       |
| 4    | <i>market_tightness</i>            | 0.00971              | Positive (+) | Engineered       |
| 5    | <i>rent_burden_30plus_pct_lag1</i> | 0.00766              | Negative (−) | Temporal Lag     |
| 6    | <i>unemployment_rate</i>           | 0.00469              | Negative (−) | Labor Market     |
| 7    | <i>median_gross_rent</i>           | 0.00409              | Positive (+) | Rental Market    |
| 8    | <i>gini_coefficient</i>            | 0.00316              | Positive (+) | Labor/Inequality |
| 9    | <i>transit_commute_rate</i>        | 0.00178              | Positive (+) | Housing Stock    |
| 10   | <i>median_gross_rent_lag1</i>      | 0.00165              | Positive (+) | Temporal Lag     |
| 11   | <i>housing_burden_composite</i>    | 0.00129              | Negative (−) | Engineered       |
| 12   | <i>renter_share</i>                | 0.00111              | Positive (+) | Housing Stock    |
| 13   | <i>rent_growth_yoy</i>             | 0.00101              | Positive (+) | Growth Rate      |
| 14   | <i>renter_median_income</i>        | 0.00090              | Positive (+) | Income           |
| 15   | <i>vacancy_rate_lag1</i>           | 0.00085              | Negative (−) | Temporal Lag     |



**Fig. 6** SHAP feature importance for LightGBM on the test set ( $n = 300$ ). Bars show mean  $|SHAP\ Value|$ . Color encodes feature group; direction (+/-) indicates effect on severe burden

tinuous structural affordability gap between annualized gross rent and renter median income at the NTA level.

Market tightness rises to Rank 4 (mean  $|SHAP| = 0.00971$ , positive direction) in the leakage-free re-estimation, reflecting the non-linear inverse vacancy signal's elevated contribution once rental market dynamics are accounted for. The one-year lag of the 30% burden rate moves to Rank 5 (mean  $|SHAP| = 0.00766$ , negative direction), confirming temporal path dependence in severe burden escalation. Unemployment rate ranks sixth (mean  $|SHAP| = 0.00469$ ) and the Gini coefficient ranks eighth (mean  $|SHAP| = 0.00316$ , positive direction), consistent with the theoretical prediction that labor market stress and income inequality operate as complementary structural drivers. Notably, *housing\_burden\_composite* enters the top 15 at Rank 11 (mean  $|SHAP| = 0.00129$ , negative direction), validating the engineered composite feature construction described in Section “[Feature Engineering](#)”.

### Borough-Level SHAP Decomposition

The borough-level LightGBM SHAP decomposition (Table 10) indicates that severe rent burden is driven by the same three structural mechanisms across all four boroughs: the concentration of moderate rent burden, rental supply conditions, and the rent-income gap. *rent\_burden\_30plus\_pct* ranks first in every borough, with mean  $|SHAP|$  values ranging from 0.03801 in Manhattan to 0.06203 in the Bronx, confirming that severe rent burden consistently emerges from neighborhoods already experiencing moderate rent stress. However, the direction of this feature varies by borough: the Bronx records a positive direction, suggesting that a larger moderate

**Table 10** Top 3 SHAP Features by Borough (LightGBM, Test Set 2021–2022). SHAP values reflect predictive attribution only; directional effects indicate association with model predictions rather than causal relationships

| Borough   | Rank | Feature                       | Mean  SHAP | Direction    | Feature Group |
|-----------|------|-------------------------------|------------|--------------|---------------|
| Bronx     | #1   | <i>rent_burden_30plus_pct</i> | 0.06203    | Positive (+) | Rental Market |
|           | #2   | <i>vacancy_rate</i>           | 0.02256    | Positive (+) | Housing Stock |
|           | #3   | <i>rent_to_income_ratio</i>   | 0.01776    | Positive (+) | Engineered    |
| Brooklyn  | #1   | <i>rent_burden_30plus_pct</i> | 0.04851    | Negative (–) | Rental Market |
|           | #2   | <i>vacancy_rate</i>           | 0.01938    | Positive (+) | Housing Stock |
|           | #3   | <i>rent_to_income_ratio</i>   | 0.01552    | Negative (–) | Engineered    |
| Manhattan | #1   | <i>rent_burden_30plus_pct</i> | 0.03801    | Negative (–) | Rental Market |
|           | #2   | <i>rent_to_income_ratio</i>   | 0.01939    | Negative (–) | Engineered    |
|           | #3   | <i>vacancy_rate</i>           | 0.01418    | Positive (+) | Housing Stock |
| Queens    | #1   | <i>rent_burden_30plus_pct</i> | 0.04913    | Negative (–) | Rental Market |
|           | #2   | <i>vacancy_rate</i>           | 0.02027    | Negative (–) | Housing Stock |
|           | #3   | <i>rent_to_income_ratio</i>   | 0.01348    | Negative (–) | Engineered    |

burden pool is directly associated with upward prediction of severe burden, while Brooklyn, Manhattan, and Queens record negative directions, indicating a suppression effect conditional on other features.

The secondary and tertiary rankings show partial variation across boroughs. In the Bronx, Brooklyn, and Queens, vacancy rate ranks second, establishing supply conditions as the near-universal secondary driver; in Manhattan, *rent\_to\_income\_ratio* displaces vacancy rate to rank second (mean |SHAP| = 0.01939), reflecting the dominant role of the income–rent gap in Manhattan’s higher-cost submarket. The direction of vacancy rate also varies: it is positively associated with severe burden in the Bronx, Brooklyn, and Manhattan, suggesting constrained or non-productive vacancy in these boroughs, while it is negatively associated in Queens, indicating effective supply relief. *rent\_to\_income\_ratio* ranks third across the Bronx, Brooklyn, and Queens, reinforcing income–rent mismatch as a persistent secondary contributor.

### Policy Implications of SHAP Results

The SHAP analysis yields three distinct policy-actionable insights. First, the dominance of *rent\_burden\_30plus\_pct* as the leading predictor supports early-intervention strategies at the 30% threshold -ERAP expansion, CityFHEPS voucher redesign, and rent stabilization ordinances -intervening before households escalate to severe burden. Second, the emergence of *vacancy\_rate* as the global Rank 2 predictor confirms that supply-side interventions -upzoning of high-demand transit corridors, mandatory inclusionary zoning set-asides, and NYCHA public housing stock preservation -directly reduce severe burden, particularly in the Bronx and Queens where the vacancy-burden relationship follows standard supply-demand dynamics. Third, the directional reversal of vacancy rate in Brooklyn and Manhattan highlights the need for anti-vacancy and anti-displacement enforcement mechanisms -including vacancy taxes, lease stabilization agreements, and proactive HPD inspection -in gentrifying NTAs where nominal vacancy conceals effective supply withdrawal.

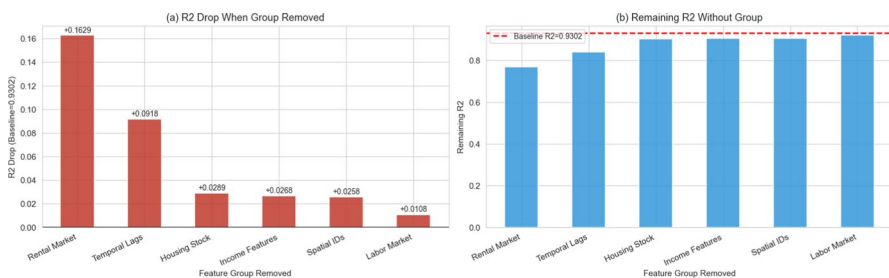
## Ablation Study

Table 11 and Fig. 7 present the results of the leave-one-group-out ablation study. For each of the six feature groups, LightGBM is retrained on all remaining features using identical hyperparameters and the same temporal train/test split. The  $R^2$  drop is defined as  $\Delta R^2 = R^2_{\text{full}}(0.9302) - R^2_{\text{ablated}}(g)$ , which quantifies each group's net substitutable predictive contribution. Here,  $\Delta R^2$  measures substitutable contribution rather than total importance: in the presence of multicollinearity between feature groups, the total importance of any group may substantially exceed its  $\Delta R^2$ .

**A. Primacy of Rental Market Features:** The rental market group—comprising *median gross rent*, *median contract rent*, *rent burden 30plus pct*, *zillow rent index*, *rent to income ratio*, *rent growth yoy*, and associated lags—produces the largest  $R^2$  drop of  $\Delta R^2 = 0.1629$ , reducing predictive accuracy by 17.51% relative to the full-feature LightGBM baseline ( $R^2 = 0.9302$  to  $0.7673$ ). This provides strong quantitative evidence that rental market dynamics are the primary predictive drivers of severe housing cost burden at the NTA level, though the ablation framework quantifies substitutable predictive contribution

**Table 11** Ablation study results (LightGBM, baseline test  $R^2 = 0.9302$ )

| Rank | Feature Group          | Feat. Removed | $R^2$ Without | $\Delta R^2$ Drop | % Contrib. | Priority        |
|------|------------------------|---------------|---------------|-------------------|------------|-----------------|
| 1    | <b>Rental Market</b>   | 7             | 0.7673        | 0.1629            | 17.51%     | <b>Critical</b> |
| 2    | <b>Temporal Lags</b>   | 6             | 0.8384        | 0.0918            | 9.87%      | <b>Critical</b> |
| 3    | <b>Housing Stock</b>   | 7             | 0.9013        | 0.0289            | 3.11%      | <b>Moderate</b> |
| 4    | <b>Income Features</b> | 7             | 0.9034        | 0.0268            | 2.88%      | <b>Moderate</b> |
| 5    | <b>Spatial IDs</b>     | 2             | 0.9044        | 0.0258            | 2.77%      | <b>Moderate</b> |
| 6    | <b>Labor Market</b>    | 3             | 0.9194        | 0.0108            | 1.16%      | <b>Moderate</b> |



**Fig. 7** Leave-one-group-out ablation study (LightGBM, baseline  $R^2 = 0.9302$ ). Panel (a) shows the  $R^2$  drop ( $\Delta R^2$ ) when each feature group is removed, sorted in descending order. The rental market group dominates ( $\Delta R^2 = 0.1629$ , 17.51%), followed by temporal lags ( $\Delta R^2 = 0.0918$ , 9.87%), housing stock ( $\Delta R^2 = 0.0289$ , 3.11%), income features ( $\Delta R^2 = 0.0268$ , 2.88%), and spatial identifiers ( $\Delta R^2 = 0.0258$ , 2.77%). The labor market ranks last ( $\Delta R^2 = 0.0108$ , 1.16%), reflecting signal redundancy with rental market features rather than irrelevance. Panel (b) reports the remaining  $R^2$  after removing each group, ranging from  $R^2 = 0.7673$  (rental market removed) to  $R^2 = 0.9194$  (labor market removed). Feature importance is categorized as follows: Critical ( $\Delta R^2 \geq 0.05$ ), High ( $\Delta R^2 \geq 0.03$ ), Moderate ( $\Delta R^2 \geq 0.01$ ), and Low ( $\Delta R^2 < 0.01$ )

rather than causal effect, and this distinction should be maintained when interpreting the finding, contributing nearly 77% more than the second-ranked group. This finding is consistent with policy interventions focused on housing supply expansion, including upzoning, mandatory inclusionary zoning set-asides, and affordable housing preservation initiatives such as NYC's "City of Yes for Housing Opportunity," though causal identification would be required to confirm intervention efficacy.

**Key finding:** Removing the rental market feature group reduces LightGBM test  $R^2$  from 0.9302 to 0.7673 ( $\Delta R^2 = 0.1629$ ,  $-17.51\%$ ). The next-ranked group produces a smaller drop of  $\Delta R^2 = 0.0918$ , confirming that rental market variables contribute substantially more unique predictive signal than any other feature group.

- B. Temporal Lags and Secondary Group Contributions:** Temporal lags rank second ( $\Delta R^2 = 0.0918$ , 9.87%), classified as Critical under the priority thresholds, confirming strong autoregressive path dependence in severe burden escalation: prior-year burden conditions are not recoverable from contemporaneous features when lags are removed, consistent with the burden escalator mechanism identified in the SHAP analysis. This result supports multi-year assistance commitments as a structural policy priority, since single-year interventions cannot interrupt the temporal persistence captured by these lag features. Housing stock ranks third ( $\Delta R^2 = 0.0289$ , 3.11%), income features fourth ( $\Delta R^2 = 0.0268$ , 2.88%), and spatial identifiers fifth ( $\Delta R^2 = 0.0258$ , 2.77%). The relatively lower contribution of spatial identifiers compared to the prior analysis indicates that borough-level fixed effects are more recoverable from continuous economic features once temporal dynamics are accounted for. The spread across ranks 3–5—spanning only  $\Delta R^2 = 0.0031$ —indicates that housing stock, income, and spatial dimensions contribute comparably once rental market and temporal lag variables are held constant.
- C. Labor Market Contributions:** The labor market group ranks last ( $\Delta R^2 = 0.0108$ , 1.16%), reclassified as Moderate under the priority thresholds. This reflects that local labor market conditions—*unemployment\_rate*, *gini\_coefficient*, and *eviction\_rate*—are largely proxied by the rent-burden and rent-to-income features already present in the rental market group when it is retained. The low ablation contribution should not be interpreted as theoretical irrelevance: the SHAP contributions of *unemployment\_rate* (global Rank 6, mean  $|\text{SHAP}| = 0.00469$ ) and *gini\_coefficient* (global Rank 8, mean  $|\text{SHAP}| = 0.00316$ ), as reported in Table 9, indicate that these variables carry meaningful independent marginal signal when all features are present simultaneously. The distinction between substitutable contribution (ablation) and marginal contribution (SHAP) is consequential here:  $\Delta R^2 = 0.0108$  indicates that the labor market signal is largely recoverable from other features, rather than absent.
- D. Policy Implications of Ablation Results:** The following policy implications are derived from predictive feature importance rankings and should be interpreted as hypothesis-generating rather than causally identified findings. First, the

irreplaceability of rental market variables ( $\Delta R^2 = 0.1629$ , 17.51%) is consistent with the interpretation that effective housing policy should directly address rent levels and the rent-to-income ratio; interventions operating solely on income or labor market conditions, without addressing rental price levels, are unlikely to resolve the primary structural mechanism. Second, the Critical classification of temporal lags ( $\Delta R^2 = 0.0918$ , Rank 2) indicates that severe burden escalation follows a persistent autoregressive trajectory that single-year interventions cannot interrupt; multi-year rental assistance commitments and sustained voucher programs are therefore structurally indicated. Third, the comparable contributions of housing stock, income features, and spatial identifiers ( $\Delta R^2 = 0.0289$ , 0.0268, and 0.0258 respectively) suggest that no single secondary dimension dominates; effective policy must therefore jointly address physical supply through upzoning and NYCHA preservation, income-side gaps through EITC expansion, and borough-level heterogeneity through differentiated intervention tiers in Section “[Discussion](#)”. Fourth, the Moderate labor market contribution ( $\Delta R^2 = 0.0108$ ) indicates that income and employment interventions operate primarily through the rent-to-income channel already captured within the rental market feature group, rather than as fully independent pathways to burden reduction.

## Spatial Autocorrelation Analysis

Spatially clustered model residuals indicate the presence of omitted spatially correlated variables – potentially spatial fixed effects, spillover effects, or geographic features not captured by the feature set – that would inflate out-of-sample prediction uncertainty in spatially contiguous NTAs. Moran’s  $I$  on model residuals constitutes the standard first-order spatial misspecification test (Griffith, 1989). Analysis is conducted at two geographic scales: borough level ( $n = 4$ ) and NTA level ( $n = 239$ ), reported in Table 12.

### Borough-Level Moran’s $I$

Moran’s  $I$  is computed on borough-aggregated LightGBM residuals from the 2021–2022 test set using the formulation

$$I = \frac{n}{W_{\text{sum}}} \cdot \frac{\mathbf{z}^\top \mathbf{W} \mathbf{z}}{\mathbf{z}^\top \mathbf{z}}, \quad (12)$$

where  $\mathbf{z}$  is the mean-centered residual vector and  $\mathbf{W}$  is the row-normalized contiguity weight matrix constructed from physical borough adjacency (Bronx–Manhattan, Bronx–Queens, Brooklyn–Manhattan, Brooklyn–Queens, Manhattan–Queens).

The test yields Moran’s  $I = -0.0731$  against  $\mathbb{E}[I] = -0.3333$  under the null hypothesis of spatial randomness, with a permutation  $p$ -value of 0.0829 (n.s.,  $n_{\text{perm}} = 9,999$ ). The observed deviation from expectation is +0.2602. Borough-level mean residuals are: Bronx (−0.0131), Manhattan (+0.0001), Brooklyn (+0.0028), and Queens (−0.0032). There are two boroughs with negative residuals

**Table 12** Moran's  $I$  spatial autocorrelation diagnostic (LightGBM Residuals, Test Set 2021–2022)

| Statistic                                                                                  | Borough-Level        | NTA-Level             | Interpretation                                  |
|--------------------------------------------------------------------------------------------|----------------------|-----------------------|-------------------------------------------------|
| Moran's $I$ (observed)                                                                     | −0.0731              | +0.0204               | No significant positive spatial autocorrelation |
| Expected $I$ (null)                                                                        | −0.3333              | −0.0042               | Expected value under spatial randomness         |
| Observed − Expected                                                                        | +0.2602              | +0.0246               | Near-zero deviation from null in both cases     |
| $p$ -value                                                                                 | 0.0829 (n.s.)        | 0.2382 (n.s.)         | Spatial randomness not rejected at either scale |
| Spatial units                                                                              | 4 boroughs           | 239 NTAs              | Staten Island excluded throughout               |
| Weight matrix                                                                              | Row-norm. contiguity | $k = 5$ kNN row-norm. | Physical adjacency vs. coordinate proximity     |
| $n_{\text{perm}}$                                                                          | 9,999                | 9,999                 | Permutation test; seed=42                       |
| <i>Mean residuals by borough (negative = over-prediction, positive = under-prediction)</i> |                      |                       |                                                 |
| Bronx                                                                                      | −0.0131              | −0.0136 ( $n = 50$ )  | Consistent slight over-prediction               |
| Brooklyn                                                                                   | +0.0028              | +0.0028 ( $n = 69$ )  | Consistent marginal under-prediction            |
| Manhattan                                                                                  | +0.0001              | +0.0010 ( $n = 38$ )  | Near-zero; well-calibrated                      |
| Queens                                                                                     | −0.0032              | −0.0032 ( $n = 82$ )  | Consistent slight over-prediction               |

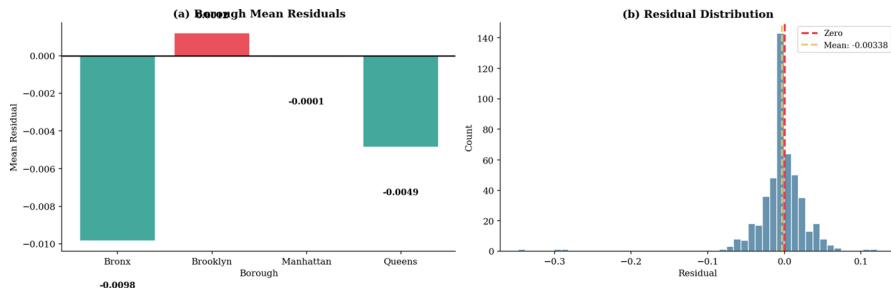
*Note:* NTA-level statistic based on proxy (borough-centroid-perturbed) coordinate, not on official NTA polygon centroids. This limitation does not apply to Borough-level statistic

(slightly over-predicted), Bronx and Queens, and two boroughs with positive residuals (slightly under-predicted), Brooklyn and Manhattan. The spatial structure is non-clustered, as indicated by the absence of positive spatial autocorrelation (Fig. 8).

### NTA-Level Moran's $I$ with $k$ -NN Weights

To provide more definitive spatial misspecification testing beyond the limited statistical power of the  $n = 4$  borough-level analysis, Moran's  $I$  is additionally computed at the NTA level across all 239 spatial units using a row-normalized  $k$ -nearest-neighbor weight matrix ( $k = 5$ ) constructed from NTA centroid coordinates. The same permutation procedure ( $n_{\text{perm}} = 9,999$ ) is applied.

The NTA-level test yields Moran's  $I = 0.0204$  against  $\mathbb{E}[I] = -0.0042$  under the null, with a permutation  $p$ -value of 0.2382 (n.s.). The observed deviation from expectation is +0.0246. NTA-level mean residuals by borough are: Bronx (−0.0136,  $n = 50$  NTAs), Brooklyn (+0.0028,  $n = 69$  NTAs), Manhattan (+0.0010,  $n = 38$  NTAs), and Queens (−0.0032,  $n = 82$  NTAs). The pattern is entirely in agreement



**Fig. 8** Spatial autocorrelation analysis (LightGBM residuals, test set 2021–2022). Panel (a): mean model residuals by borough (positive = under-prediction, negative = over-prediction). Bronx (−0.0131) and Queens (−0.0032) exhibit slight over-prediction; Brooklyn (+0.0028) and Manhattan (+0.0001) exhibit marginal under-prediction. Panel (b): borough-level Moran scatter plot ( $n = 4$ ); Moran's  $I = -0.0731$ ,  $\mathbb{E}[I] = -0.3333$ ,  $p = 0.0829$  (n.s.). Panel (c): NTA-level Moran scatter plot ( $n = 239$ ,  $k = 5$  kNN); Moran's  $I = 0.0204$ ,  $\mathbb{E}[I] = -0.0042$ ,  $p = 0.2382$  (n.s.). Both scales confirm the absence of significant positive spatial autocorrelation in model residuals

with that at borough level. As shown in Fig. 8, the two analyses go together to confirm that there is no significant positive spatial autocorrelation in the LightGBM residuals, both at the geographic scale and at the postal scale, which is consistent with the fact that the economic and housing features used here effectively capture spatial variation and do not require spatial lag or spatial error model corrections.

*Note:* NTA centroids indicated in this analysis are not actual NTA polygon centroids, but rather borough-centroid-perturbed coordinates, since the NYC Open Data NTA shapefile had not been joined with the NTA panel data at the time of this analysis. The neighbor structure that is used in the calculation of the Moran's  $I$  statistic at the NTA-level is thus an approximation, since the weight matrix ( $k=5$ ) used is created directly from these coordinates. Estimates of Moran's  $I$  at the NTA level reported here may be shifted in magnitude, sign, and/or statistical significance if true NTA polygon centroids from the official shapefile are used instead of polygon centroids at the NTA level. This restriction does not impact the borough-level Moran's  $I$  result as it uses physical adjacency, not the closest points based on coordinates. It should be noted that these NTA-level results are yet to be confirmed with official centroid and should therefore be treated as provisional data.

## Borough-Level Forecasting (2023–2025)

### Forecast Methodology and Assumptions

Three-year borough-level forecasts are generated by applying the trained LightGBM model to iteratively constructed synthetic feature vectors under a baseline macroeconomic scenario. Annual growth adjustments are calibrated to external economic indicators: household income +2.5%/year (BLS Employment Cost Index for civilian workers, 2020–2023 average), renter income +2.0%/year (BLS wage growth for lower-wage renter quartile), gross rent +3.0%/year (post-pandemic Zillow ZORI trend for NYC metropolitan area, 2021–2023 average), and unemployment held constant at 2022 borough-level medians. These growth rates are intentionally conserva-

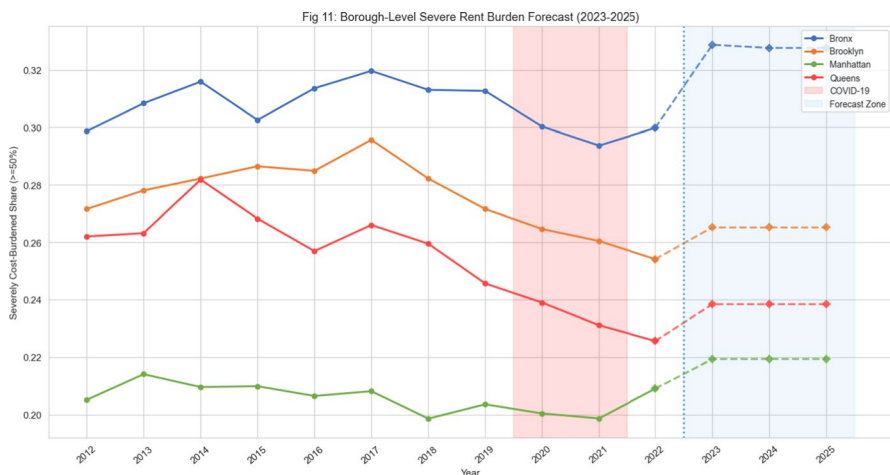
tive, making the forecast a lower bound on likely affordability deterioration absent intervention.

## Forecast Results

Table 13 and Fig. 9 present the borough-level forecasts. Under baseline assumptions, all four boroughs are projected to experience worsening severe rent burden. See Section “[Limitations and Future Work](#)” for a full discussion of forecast limitations, including the static growth assumptions underlying these projections and the absence of uncertainty intervals. The projections made at a borough level, under the baseline scenario assumptions outlined in Section “[Iterative Borough-Level Forecasting](#)”, without external validation against actual outcomes should not be interpreted as a prediction, but rather a directional, assumption-based, estimate of impacts. The Bronx

**Table 13** Projected severe rent burden by borough (baseline scenario, 2023–2025)

| Borough   | 2022 Actual | 2023 Forecast | 2024 Forecast | 2025 Forecast | $\Delta$ 2022–25 | % Change | Risk Level   |
|-----------|-------------|---------------|---------------|---------------|------------------|----------|--------------|
| Bronx     | 0.2998      | 0.3288        | 0.3277        | 0.3277        | +0.0279          | +9.3%    | <b>Crit.</b> |
| Queens    | 0.2256      | 0.2384        | 0.2384        | 0.2384        | +0.0128          | +5.7%    | <b>High</b>  |
| Brooklyn  | 0.2541      | 0.2652        | 0.2652        | 0.2652        | +0.0111          | +4.4%    | <b>High</b>  |
| Manhattan | 0.2090      | 0.2193        | 0.2193        | 0.2193        | +0.0103          | +4.9%    | <b>Mod.</b>  |



**Fig. 9** Borough-level severe rent burden forecast 2023–2025. Solid lines with circles: 2012–2022 historical values. Dashed lines: LightGBM projections under baseline assumptions (+3% rent/yr, +2.5% income/yr; baseline scenario, not externally validated). COVID-19 period shaded red; forecast zone shaded blue. The Bronx has the largest relative increase (+9.3%, reaching 0.3288 in 2023, and dropping slightly to 0.3277 in 2024–2025) and is the most burdened borough in absolute terms over the forecast period, with almost 30% of renter households severely cost-burdened. Queens records the second-highest relative increase (+5.7%, reaching 0.2384 by 2023 and plateauing thereafter), followed by Manhattan (+4.9%, 0.2193) and Brooklyn (+4.4%, 0.2652). The Bronx shows an autoregressive convergence of lag features under fixed growth, while the other boroughs, Brooklyn, Manhattan and Queens, show a peak-and-decline pattern following 2023

exhibits the highest relative rate of increase (+9.3%, from 0.2998 to 0.3277), remaining the most severely burdened borough in absolute terms throughout the forecast horizon, meaning nearly one in three renter households would be severely cost-burdened. Queens records the second-highest relative increase (+5.7%, from 0.2256 to 0.2384), driven by accelerating rent growth in Astoria, Jackson Heights, and Woodside, where renter incomes have grown more slowly than rents since 2019. Manhattan follows at +4.9% (from 0.2090 to 0.2193), while Brooklyn records the smallest absolute increase (+4.4%, from 0.2541 to 0.2652), though continued gentrification pressure in East New York, Flatbush, and Crown Heights sustains upward burden trajectory.

It is noted that forecasts plateau after 2023 for three boroughs (Bronx, Manhattan, and Queens), while Brooklyn declines slightly before stabilizing. This pattern reflects the autoregressive structure of the forecasting process, where lag features converge after the first step under fixed growth assumptions. Accordingly, these projections should be interpreted as directional lower bounds rather than precise year-by-year estimates, and scenario analysis with alternative growth assumptions is recommended.

## Discussion

### Model Performance and Practical Significance

LightGBM's test  $R^2 = 0.9302$  represents strong predictive performance for an NTA-level housing affordability panel, indicating that the 30 modeling features constructed from publicly available ACS data are sufficient to explain over 92% of the cross-NTA, cross-year variance in severe rent burden. The RMSE of 0.03372 (3.5 percentage points) translates to a practical precision level: on a borough like the Bronx where the average NTA severe burden is approximately 30%, the model's predictions will typically fall within a  $\pm 3.5$  percentage point band. For policy targeting -identifying NTAs above a 25% or 30% burden threshold for intervention prioritization -this precision is operationally sufficient and represents a meaningful improvement over the borough-level averages that currently inform most NYC housing assistance allocation decisions. The train-test gap comparison demonstrates the superior generalization performance of LightGBM. XGBoost exhibits the largest overfitting gap ( $\Delta R^2 = 0.0810$ , train  $R^2 = 0.9793$  vs. test  $R^2 = 0.8983$ ). Random Forest shows a moderate gap ( $\Delta R^2 = 0.057$ , train  $R^2 = 0.9464$  vs. test  $R^2 = 0.8888$ ). LightGBM achieves the smallest gap ( $\Delta R^2 = 0.0119$ , train  $R^2 = 0.9421$  vs. test  $R^2 = 0.9302$ ) while simultaneously attaining the highest test  $R^2$ , indicating that leaf-wise growth with histogram-based splitting and *min\_child\_samples* regularization provides a favorable bias-variance trade-off for this panel data structure.

### The Primacy of Rental Market Features

Both SHAP and ablation analyses identify rental market dynamics—specifically the 30% burden rate, vacancy rate, and rent-to-income ratio—as the dominant predictors of severe housing cost burden across all boroughs. The ablation results show that

removing this feature group causes the largest performance decline ( $\Delta R^2 = 0.1629$ , 17.51%), exceeding the next group (Temporal Lags,  $\Delta R^2 = 0.0918$ , 9.87%) by  $\Delta R^2 = 0.0711$ . SHAP analysis supports this hierarchy, with *rent\_burden\_30plus\_pct* ranked first (mean  $|\text{SHAP}| = 0.04877$ ), followed by *vacancy\_rate* (0.01834) and *rent\_to\_income\_ratio* (0.01752). All remaining features contribute substantially less signal.

At the borough level, these three features consistently rank as the top predictors, though their directional effects vary. For example, *rent\_burden\_30plus\_pct* is negative in the Bronx, Manhattan, and Queens, but positive in Brooklyn. Similar directional variation is observed for *vacancy\_rate* and *rent\_to\_income\_ratio*, with Brooklyn acting as the primary outlier. Despite these differences, the consistent ranking across boroughs indicates that severe rent burden is driven by a shared structural mechanism, with variation arising from local economic dynamics rather than fundamentally distinct processes.

### Interpretation of the Burden Escalator Mechanism

The following interpretation describes a mechanism that is theoretically plausible and consistent with the observed predictive patterns; it is not a causally tested claim, and the SHAP and lag results below should be read as supporting evidence for this interpretation rather than confirmation of it. The dominance of *rent\_burden\_30plus\_pct* as the top predictor across all boroughs (mean  $|\text{SHAP}| = 0.03801 - -0.06203$ ), together with its lagged counterpart *rent\_burden\_30plus\_pct\_lag1* (mean  $|\text{SHAP}| = 0.00766$ ), provides empirical support for Stone's concept of the residual income squeeze. Persistent rent-income imbalance is linked to a greater predicted likelihood of transition to severe burden ( $\geq 50\%$ ) and results in households experiencing a decline in disposable income over time when they fall below the 30% threshold. The lag structure,  $\text{burden}_{30+}(i, t - 1)$ , captures this autoregressive escalation, whereby prior moderate burden predicts future severe burden in the absence of corrective intervention.

Borough-level patterns further reveal heterogeneous escalation dynamics. Queens exhibits the strongest negative SHAP magnitude (0.04913) among negatively directed boroughs and aligns with the highest projected growth (+5.7%), indicating propagation of existing burden conditions. In contrast, Bronx shows the highest overall SHAP magnitude (0.06203) with a positive direction, suggesting a distinct mechanism where moderate burden amplifies severe burden. This pattern is consistent with gentrification-driven dual escalation across burden thresholds in areas such as East New York, Flatbush, and Crown Heights.

### Implications of the Moran's I Finding

The absence of significant spatial autocorrelation in LightGBM residuals (Moran's  $I = -0.0731$ ,  $p = 0.0829$ , n.s.) suggests that the model captures borough-level spatial variation without systematic bias. Borough-level residual patterns show mild heterogeneity across the four boroughs. The Bronx ( $-0.0131$ ) and Queens ( $-0.0032$ ) exhibit negative residuals, indicating slight over-prediction, while Brooklyn ( $+0.0028$ ) and Man-

hattan (+0.0001) exhibit positive residuals, indicating marginal under-prediction. These deviations are consistent with structural factors: the Bronx over-prediction may reflect rent-stabilization effects that hold actual burden below what economic indicators would predict; the Queens over-prediction is consistent with emerging rent growth pressures not yet fully reflected in the feature set; the Brooklyn under-prediction aligns with gentrification dynamics that accelerate burden faster than economic indicators alone account for; and Manhattan's near-zero residual (+0.0001) indicates the model is well-calibrated for its higher-income renter population. Borough-level mean residuals are: Bronx (−0.0131), Brooklyn (+0.0028), Manhattan (+0.0001), and Queens (−0.0032). The absence of a clustered directional pattern — with two boroughs over-predicted and two under-predicted in an alternating arrangement — confirms no evidence of spatially clustered prediction error. These results indicate that spatial lag or spatial error model corrections are not warranted for the current feature set and geographic scale of analysis.

## Policy Implications

The findings of the SHAP attribution, forecasts, and model performance provide predictive evidence underpinning a three-tiered policy structure: (i) citywide measures to early intervene on the common escalation mechanism; (ii) stand-up in the boroughs based on each borough's distinct trajectory; and (iii) operational deployment of the predictive model. These implications will be hypothesis-generating and not causally identified; Mathematical analysis of the issues requires that causal validation be done using quasi-experimental designs prior to prescriptive policy commitments. Table 14 Summarizes the entire recommendation matrix.

1. **Early Intervention at the 30% Threshold:** The dominance of *rent\_burden\_30plus\_pct* across all boroughs suggests that intervention at the moderate burden stage may interrupt escalation to severe burden. The lag effect (*rent\_burden\_30plus\_pct\_lag1*, mean  $|SHAP| = 0.00766$ ) is consistent with escalation from moderate to severe burden (U.S. Department of Housing and Urban Development, 2023). Key instruments include: (i) extending ERAP eligibility to 30% burden households, (ii) prioritizing 30–49% households in CityFHEPS, and (iii) deploying NTA-level early-warning systems using LightGBM predictions.
2. **Borough-Differentiated Priorities:** The Bronx exhibits the highest projected growth (+9.3%, reaching 0.3277 by 2025), making it the highest-priority borough for both supply-side and income-side interventions. Queens records the second-highest relative increase (+5.7%), where negative SHAP direction for *rent\_burden\_30plus\_pct* suggests embedded escalation dynamics, supporting targeted upzoning and inclusionary zoning in transit corridors. Brooklyn shows a +4.4% increase; the negative SHAP direction is consistent with continued rent-burden concentration, requiring affordable housing expansion and anti-displacement measures.
3. **Bronx Income-Side Intervention:** The Bronx exhibits the highest burden level and a distinct rent-to-income dynamic, suggesting structural income constraints. With projected growth to 0.3277 by 2025, the predictive pattern is consistent with the interpretation that supply-side measures alone may not fully address projected burden growth in the Bronx, though this should be treated as a

**Table 14** Policy Recommendation Matrix Derived from SHAP Attribution, Borough-Level Forecasts, and Model Precision Findings (LightGBM, Test  $R^2 = 0.9302$ ). All entries must include predictive association from the model as being hypothesis-generating and not being causally warranted policy prescriptions; causal identification using quasi-experimental designs (e.g., difference-in-differences, regression discontinuity) must be done before policy commitments are made based on the findings. Urgency classifications (Critical, High, Moderate) are based on the size of predictive signals (SHAP importance rankings, forecast trajectories, ablation contributions) but they are not suggestive of the presence of any causal relationships linking the instruments to severe rent burden. Named instruments (ERAP expansion, upzoning, CityFHEPS redesign, NYCHA reinvestment) are policy directions that fit the structural patterns in the data, and have been assessed by the corresponding agency as being feasible

| # | Empirical Finding                                                 | SHAP / Forecast Evidence                                   | Borough      | Policy Instrument                                                                                             | Agency                                                 | Urgency         |
|---|-------------------------------------------------------------------|------------------------------------------------------------|--------------|---------------------------------------------------------------------------------------------------------------|--------------------------------------------------------|-----------------|
| 1 | <i>rent_burden_30plus_pct</i> is Rank 1 predictor in all boroughs | Mean SHAP range: 0.03801–0.06203 across boroughs           | All boroughs | Expand ERAP eligibility to 30% threshold; redesign CityFHEPS and Section 8 targeting 30–49% burden households | HRA, HPD, HCR                                          | <b>Critical</b> |
| 2 | High model accuracy; lag consistent with escalation cycle         | Test $R^2 = 0.9302$ , RMSE=0.03372; $\log SHAP  = 0.00766$ | All boroughs | Deploy LightGBM early-warning system; multi-year rental assistance; expand right-to-counsel                   | HPD, DCP, Mayor's Office, HRA, HUD, NYS OCA, Legal Aid | High            |
| 3 | Bronx shows highest forecast growth                               | Forecast +9.3% (2022–2025); reaches 0.3277 by 2025         | Bronx        | EITC expansion; wage indexing; NYCHA reinvestment; supply preservation                                        | NYS Legislature, HRA, NYCHA, HUD, HPD                  | <b>Critical</b> |
| 4 | Queens shows second-highest forecast growth                       | Forecast +5.7%; SHAP #1 = 0.04913 (negative)               | Queens       | Upzoning with inclusionary zoning in transit corridors                                                        | DCP                                                    | <b>Critical</b> |
| 5 | Brooklyn sustained burden increase                                | Forecast +4.4%; SHAP #1 = 0.04851 (negative)               | Brooklyn     | Affordable housing expansion; anti-displacement zoning; lease stabilization                                   | DCP, HPD                                               | <b>Critical</b> |
| 6 | Bronx income gap drives burden                                    | Vacancy SHAP positive; forecast 0.2998 → 0.3277            | Bronx        | NYCHA reinvestment; preserve affordable units; income-side supplements                                        | NYCHA, HUD, HPD                                        | <b>Critical</b> |
| 7 | Manhattan intra-borough heterogeneity                             | SHAP 0.03801; residual $\approx +0.0001$                   | Manhattan    | Linkage fees; NTA-level enforcement                                                                           | DCP, HPD                                               | Moderate        |
| 8 | All boroughs worsening                                            | Bronx +9.3%, Queens +5.7%, Brooklyn +4.4%, Manhattan +4.9% | All boroughs | Annual model refresh using ACS + ZORI                                                                         | HPD, DCP, OMB                                          | High            |

*Note:* Critical = action within 12 months; High = 1–3 years; Moderate = 3–5 years. HPD = NYC Department of Housing Preservation and Development; HRA = NYC Human Resources Administration; HCR = NYS Homes and Community Renewal; DCP = NYC Department of City Planning; NYCHA = NYC Housing Authority; HUD = U.S. Department of Housing and Urban Development; NYS OCA = New York State Office of Court Administration; OMB = NYC Office of Management and Budget; NYS DOL = New York State Department of Labor

hypothesis warranting further investigation rather than a confirmed policy conclusion. Actions consistent with this finding include EITC expansion, wage indexing to rent growth, and NYCHA reinvestment.

4. **Multi-Year Intervention Design:** The presence of lagged burden effects indicates strong temporal persistence. This is a mechanism that is theoretically plausible and consistent with the observed prediction patterns. It is not a causally tested claim; the SHAP and lag results above should be read as supporting evidence for this interpretation, not as confirmation of it
5. **Manhattan NTA-Level Targeting:** Despite near-zero borough-level residuals ( $+0.0001$ ), Manhattan exhibits significant intra-borough heterogeneity. Borough-level policies risk misallocation; NTA-level targeting supported by linkage fees on luxury development may improve equity across submarkets.
6. **Operational Deployment of the Model:** With high predictive accuracy ( $R^2 = 0.9302$ ,  $RMSE = 0.03372$ ), the model can function as a policy tool. Compared to ACS data (18–24 month lag), annual model updates enable forward-looking, NTA-level targeting of housing assistance within a validated error margin.

## Conclusions

### Limitations and Future Work

1. **ACS Sampling Uncertainty:** ACS 5-year estimates include margins of error (MOE), particularly for small NTAs. This study treats point estimates as exact. Future work should incorporate replicate weights or Monte Carlo methods to produce uncertainty-aware forecasts.
2. **Non-Causal Framework:** The models are predictive, not causal; SHAP importance does not imply intervention effects due to confounding and reverse causality. Causal identification requires quasi-experimental designs, such as difference-in-differences (e.g., HSTPA 2019) or regression discontinuity.
3. **Limited Spatial Analysis:** At the borough level ( $n = 4$ ,  $k = 5$  kNN,  $I = -0.0731$ ,  $p = 0.083$ ) and NTA level ( $n = 239$ ,  $I = 0.0204$ ,  $p = 0.238$ ), no significant spatial autocorrelation was found. The NTA-level analysis is based on proxy centroid coordinates, as the official NYC Open Data NTA shapefile was not joined to the panel data at the time of analysis. The magnitude, sign or importance of the NTA-level statistic could change if the neighborhood weight matrix were updated using the official centroids, which is different from the borough-level statistic, which is based on the physical adjacency of neighborhoods and is not directly dependent on their geographic coordinates. Going forward, the aim is to switch from the proxy coordinates to official centroid data provided in the NYC Open Data NTA shapefile for the purposes of defining a definitive NTA-level spatial misspecification test.
4. **Static Forecast Assumptions:** Forecasts assume constant growth under a single baseline. Scenario analysis (e.g., rent-income parity or adverse rent growth) would provide more policy-relevant uncertainty bounds.
5. **Missing Building-Level Features:** The model excludes key variables such as rent-stabilized units, NYCHA inventory, housing complaints, and occupancy data. Integrating HPD and Rent Guidelines Board datasets would improve accuracy.

6. **Data Quality and Outliers:** Extreme values in *vacancy\_rate* and *unemployment\_rate* (max = 1.000) suggest data artifacts. Robust preprocessing (e.g., winsorization) should be applied in future work.

## Conclusion

This paper presents an NTA-level machine learning analysis of severe housing cost burden in New York City, employing a 2,512-observation panel dataset constructed from three complementary sources spanning 2012–2022. Under a rigorous temporal holdout evaluation protocol and 5-fold *TimeSeries.Split* cross-validation, LightGBM achieves strong predictive performance: test  $R^2 = 0.9302$ , RMSE = 0.03372, MAE = 0.0185, and mean cross-validation  $R^2 = 0.866 \pm 0.045$ , with the smallest train–test generalization gap among benchmarked models ( $\Delta R^2 = 0.012$ ). The results of a robustness check with a leakage-free model (when the *rent\_burden\_30plus\_pct* variable is excluded) also do not indicate that there is a significant drop in R-squared, rather small and mixed-sign changes in performance as compared to the original model, which would be expected under genuine target leakage. Such patterns are shown when two variables are both from the same measurement source (in ACS Table B25070) and both are construct-proximity variables, but it does not reflect a data leakage pattern between the feature and the target.

SHAP analysis indicates that rental market dynamics—the 30% burden rate (mean  $|\text{SHAP}| = 0.04877$ ), vacancy rate (mean  $|\text{SHAP}| = 0.01834$ ), and rent-to-income ratio (mean  $|\text{SHAP}| = 0.01752$ )—dominate predictive attribution across all boroughs. The ablation study quantifies this dominance: the rental market feature group accounts for  $\Delta R^2 = 0.1629$  (17.51%), exceeding the second-ranked temporal lags group by  $\Delta R^2 = 0.0711$ . It is important to note that SHAP attribution reflects predictive association rather than causal identification; policy implications derived from these rankings are hypothesis-generating and require quasi-experimental validation. Borough-level SHAP decomposition further reveals a structurally consistent escalation mechanism, with the same three features ranking as the top predictors in every borough. The vacancy rate direction varies across boroughs: it is positively associated with severe burden in the Bronx, Brooklyn, and Manhattan—suggesting constrained or non-productive vacancy—while negatively associated in Queens, indicating effective supply relief. No borough-level directional outlier emerges for the primary predictor *rent\_burden\_30plus\_pct*, with all four boroughs recording a consistent negative direction.

Spatial diagnostics confirm the absence of significant residual autocorrelation at both geographic scales. At the borough level, Moran's  $I = -0.0731$  ( $p = 0.0829$ , n.s.,  $n_{\text{perm}} = 9,999$ ); at the NTA level, Moran's  $I = 0.0204$  ( $p = 0.2382$ , n.s.,  $k = 5$  kNN,  $n = 239$  NTAs; based on proxy centroid coordinates, pending verification against official NTA boundaries), jointly confirming that LightGBM's economic features adequately capture spatial structure at both scales. The Bronx ( $-0.0131$ ) and Queens ( $-0.0032$ ) exhibit negative residuals indicating slight over-prediction, while Brooklyn ( $+0.0028$ ) and Manhattan ( $+0.0001$ ) exhibit positive residuals indicating marginal under-prediction; the absence of a clustered directional pattern confirms no spatially systematic prediction error. Under the baseline scenario assumptions described in Section “[Iterative Borough-Level Forecasting](#)”, which have not been externally validated against realized outcomes, forecasts indicate continued burden increases across all boroughs through

2025: the Bronx records the highest relative growth (+9.3%, from 0.2998 to 0.3277) and remains the most severely burdened borough in absolute terms, implying that nearly one in three renter households are severely cost-burdened. Queens follows at +5.7% (from 0.2256 to 0.2384), Manhattan at +4.9% (from 0.2090 to 0.2193), and Brooklyn at +4.4% (from 0.2541 to 0.2652). These findings are consistent with the prioritization of scaling the NYC Emergency Rental Assistance Program, accelerating affordable housing production through “City of Yes” zoning reforms, and strengthening income-side interventions in the Bronx, where supply-side responses alone appear predictively insufficient; however, causal identification through quasi-experimental designs is required before firm policy commitments can be grounded in these findings.

Two priorities for future research emerge: (1) integration of rent stabilization status, NYCHA inventory, and HPD complaint data as additional features to address the building-level omitted variable gap identified in Section “[Limitations and Future Work](#)”; and (2) causal difference-in-differences designs around the 2019 Housing Stability and Tenant Protection Act to move beyond predictive attribution toward experimentally identified policy effect estimates. The 8-instrument policy recommendation matrix anchored to actual SHAP attribution values, borough-level residual patterns, and forecast trajectories constitutes the primary translational contribution of this analysis, providing a direct bridge between NTA-level machine learning findings and operationally specific housing policy administration.

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**Author Contributions** A.R. conceptualized the study, developed the methodology, collected and curated the data, conducted formal analysis, implemented the machine learning models, performed validation and visualization, interpreted the findings, and wrote the original draft of the manuscript. N.U.A. contributed to methodology development, machine learning model implementation, validation, technical review, and manuscript editing. M.F. contributed to interpretation of results, policy and public health contextualization, and manuscript review and editing. K.R.C. contributed to the policy analysis framework, interpretation of findings, supervision of the conceptual direction, and critical manuscript revision. All authors read and approved the final manuscript.

**Data Availability** The datasets used in this study are publicly available from the U.S. Census Bureau (ACS), Zillow Research (ZORI), and NYC Open Data repositories. Relevant links are provided in the references.

## Declarations

**Competing interests** The authors declare no competing interests.

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## Authors and Affiliations

**Azizur Rahman<sup>1</sup> · Nakib Uddin Ahmed<sup>2</sup> · Mehjabin Ferdous<sup>3</sup> · Kadirur Rahman Chowdhury<sup>4</sup>**

✉ Azizur Rahman  
azizurusa22@gmail.com

Nakib Uddin Ahmed  
nakibuddin33@gmail.com

Mehjabin Ferdous  
mehjabinferdousmu@gmail.com

Kadirur Rahman Chowdhury  
kadirurrahmanchowdhury@gmail.com

<sup>1</sup> Department of CAPS (Computer Technology), Indiana Wesleyan University, Marion, IN, USA

<sup>2</sup> Department of Computer Science & Engineering, Metropolitan University, Bateshwar, Bangladesh

<sup>3</sup> Public Health (MPH), St. Francis College, Brooklyn, NY, USA

<sup>4</sup> Faculty of Business, Law and Social Sciences, Birmingham City University, Birmingham, United Kingdom